Finite Difference Methods for the One-Way Wave Equation

Consistency, convergence, and stability under parameter sweeps

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Problem and Exact Solution

We solve the initial-boundary value problem

$$u_t + u_x = 0, \qquad x > -2, \quad t > 0, \qquad u(-2, t) = 0, \tag{1}$$

with compactly supported initial data

$$u_0(x) = \begin{cases} (1 - |x|)^2, & |x| \leq 1, \\ 0, & |x| > 1. \end{cases} \tag{2}$$

Since the speed is $a = 1$, characteristics satisfy $x - t = \mathrm{constant}$, so

$$u(x, t) = \begin{cases} u_0(x - t), & x - t \geq -2, \\ 0, & x - t < -2. \end{cases} \tag{3}$$

The left boundary is inflow; the right boundary $x = b$ is outflow.

Grid, Errors, and Sweep Parameters

Let

$$x_m = -2 + mh, \qquad t_n = nk, \qquad v_m^n \approx u(x_m, t_n), \qquad \lambda = \frac{k}{h}. \tag{4}$$

Since the wave speed is $a = 1$, λ is the Courant number. Thus $\lambda = 1$ means that the exact wave travels one grid cell during one time step.
At a final time T, define

$$E_1 = h \sum_m |v_m^N - u(x_m, T)|, \qquad E_2 = \left(h \sum_m |v_m^N - u(x_m, T)|^2 \right)^{1/2}, \tag{5}$$

$$E_\infty = \max_m |v_m^N - u(x_m, T)|, \qquad \mathrm{EOC} = \frac{\log(E_h/E_{h/2})}{\log 2}. \tag{6}$$

Task sweep

$$b \in \{3, 4, 5\}, \qquad h \in \{0.1, 0.05, 0.01\}, \qquad \lambda \in \{0.8, 0.9, 1.0, 1.01, 1.05, 1.1\}.$$

Explicit Schemes

With $\nu = ak/h = \lambda$ for this problem, the main explicit updates are

$$\text{FTFS:} \quad v_m^{n+1} = (1+\nu)v_m^n - \nu v_{m+1}^n, \tag{7}$$

$$\text{FTBS:} \quad v_m^{n+1} = (1-\nu)v_m^n + \nu v_{m-1}^n, \tag{8}$$

$$\text{FTCS:} \quad v_m^{n+1} = v_m^n - \frac{\nu}{2}(v_{m+1}^n - v_{m-1}^n), \tag{9}$$

$$\text{CTCS:} \quad v_m^{n+1} = v_m^{n-1} - \nu(v_{m+1}^n - v_{m-1}^n), \tag{10}$$

$$\text{Lax–Friedrichs:} \quad v_m^{n+1} = \frac{1+\nu}{2}v_{m-1}^n + \frac{1-\nu}{2}v_{m+1}^n, \tag{11}$$

$$\text{Lax–Wendroff:} \quad v_m^{n+1} = v_m^n - \frac{\nu}{2}(v_{m+1}^n - v_{m-1}^n) + \frac{\nu^2}{2}(v_{m+1}^n - 2v_m^n + v_{m-1}^n). \tag{12}$$

Beam–Warming uses a one-sided second-order upwind stencil:

$$v_m^{n+1} = v_m^n - \frac{\nu}{2}(3v_m^n - 4v_{m-1}^n + v_{m-2}^n) + \frac{\nu^2}{2}(v_m^n - 2v_{m-1}^n + v_{m-2}^n).$$

Implicit Schemes

The implicit methods solve a tridiagonal linear system at each time step.

$$\text{BTFS:} \quad (1-\nu)v_m^{n+1} + \nu v_{m+1}^{n+1} = v_m^n, \tag{13}$$

$$\text{BTBS:} \quad (1+\nu)v_m^{n+1} - \nu v_{m-1}^{n+1} = v_m^n, \tag{14}$$

$$\text{BTCS:} \quad v_m^{n+1} + \frac{\nu}{2}(v_{m+1}^{n+1} - v_{m-1}^{n+1}) = v_m^n, \tag{15}$$

$$\text{Crank–Nicolson:} \quad v_m^{n+1} + \frac{\nu}{4}(v_{m+1}^{n+1} - v_{m-1}^{n+1}) = v_m^n - \frac{\nu}{4}(v_{m+1}^n - v_{m-1}^n). \tag{16}$$

BTBS is the implicit upwind scheme for right-moving advection. Crank–Nicolson is formally second-order and nondissipative, so it preserves amplitudes well but may retain oscillations near nonsmooth corners.

Formal Accuracy

Taylor expansion gives the formal local truncation errors for smooth solutions.

Scheme	Local truncation error	Order if $k = O(h)$
FTFS, FTBS	$O(k + h)$	first
FTCS	$O(k + h^2)$	first
CTCS	$O(k^2 + h^2)$	second
Lax–Friedrichs	$O(k + h^2/k)$	first
Lax–Wendroff	$O(k^2 + h^2)$	second
Beam–Warming	$O(k^2 + h^2)$	second
BTFS, BTBS	$O(k + h)$	first
BTCS	$O(k + h^2)$	first
Crank–Nicolson	$O(k^2 + h^2)$	second

Important caveat

The assigned u_0 is continuous but has a corner at $x = 0$. Therefore max-norm convergence can be lower than the formal smooth-solution order, especially for higher-order dispersive schemes.

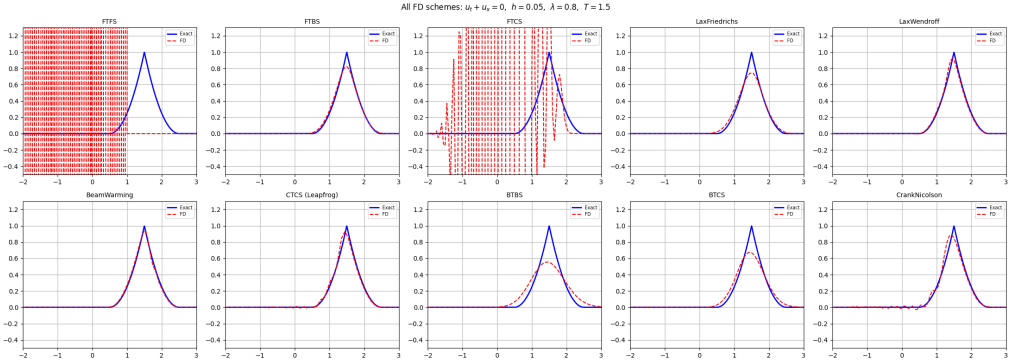
Von Neumann Stability

Insert the Fourier mode $v_m^n = G^n e^{\mathrm{i}m\theta}$ and require $|G(\theta)| \leq 1$.

Scheme	Amplification factor	Stable range for $a > 0$
FTFS	$1 + \nu - \nu e^{\mathrm{i}\theta}$	unstable
FTBS	$1 - \nu + \nu e^{-\mathrm{i}\theta}$	$0 \leq \nu \leq 1$
FTCS	$1 - \mathrm{i}\nu \sin\theta$	unstable
CTCS	$G^2 + 2\mathrm{i}\nu \sin\theta\, G - 1 = 0$	$\lvert\nu\rvert \leq 1$
Lax–Friedrichs	$\cos\theta - \mathrm{i}\nu \sin\theta$	$\lvert\nu\rvert \leq 1$
Lax–Wendroff	$1 - \mathrm{i}\nu \sin\theta + \nu^2(\cos\theta - 1)$	$\lvert\nu\rvert \leq 1$
Beam–Warming	second-order upwind symbol	$0 \leq \nu \leq 2$
BTBS	$(1 + \nu - \nu e^{-\mathrm{i}\theta})^{-1}$	all $\nu \geq 0$
BTCS	$(1 + \mathrm{i}\nu \sin\theta)^{-1}$	all ν
Crank–Nicolson	$\dfrac{1 - \frac{\mathrm{i}\nu}{2} \sin\theta}{1 + \frac{\mathrm{i}\nu}{2} \sin\theta}$	all ν

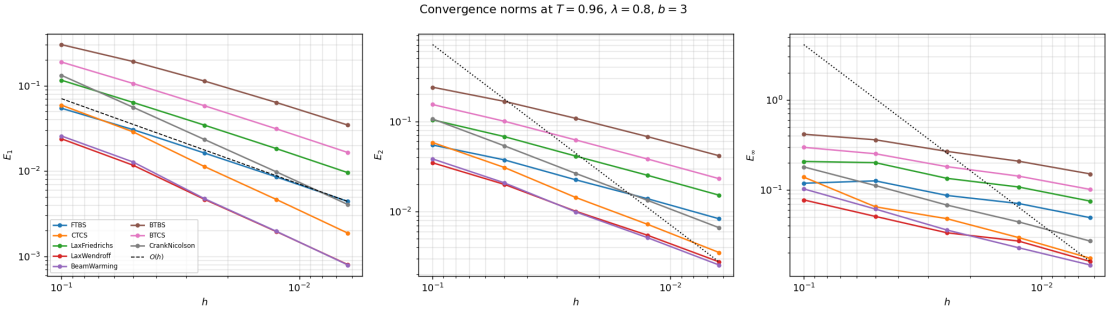
The experiments below test these theoretical predictions numerically.

Profile Comparison at $\lambda = 0.8$



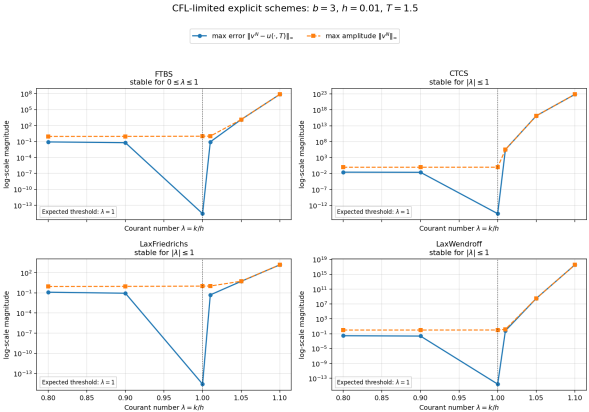
At $T = 1.5$, $h = 0.05$, and $b = 3$, the exact pulse has moved to roughly $[0.5, 2.5]$. FTBS and Lax–Friedrichs are stable but diffusive; Lax–Wendroff, CTCS, and Beam–Warming are sharper but more oscillatory near nonsmooth features.

Numerical Convergence Plots



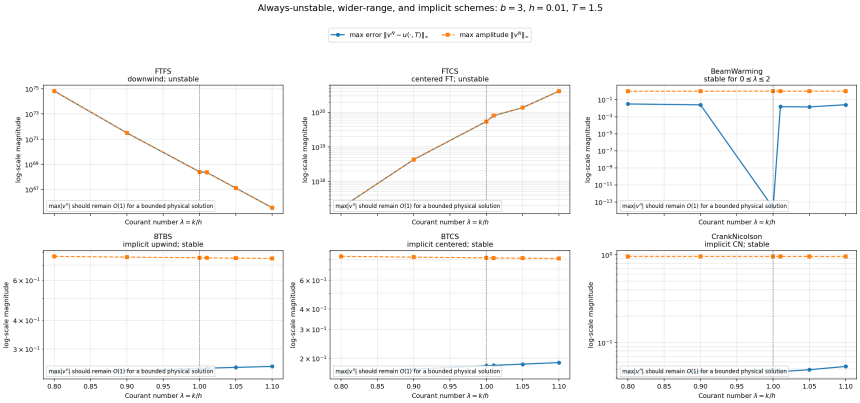
The convergence test uses $T = 0.96$, $\lambda = 0.8$, $b = 3$, and $h = 0.1, \ldots, 0.00625$. First-order methods show roughly first-order behavior in E_1; formally second-order methods have smaller errors, but the nonsmooth initial corner reduces the observed E_∞ rates. On the finest grid, representative E_∞ values are about 4.90×10^{-2} for FTBS, 1.60×10^{-2} for Lax–Wendroff, and 2.69×10^{-2} for Crank–Nicolson.

Sweep in λ: CFL-Limited Explicit Schemes



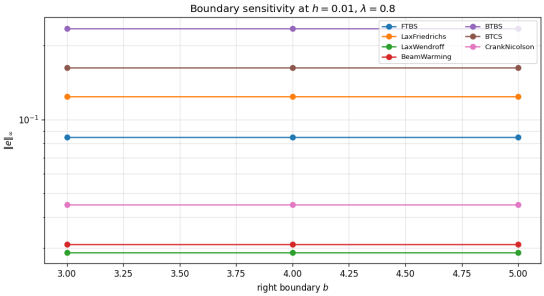
These methods are predicted by von Neumann analysis to be stable only up to the CFL threshold $\lambda = 1$. At $b = 3$, $h = 0.01$, and $T = 1.5$, the plotted quantities are the maximum error $\|v^N - u(\cdot, T)\|_\infty$ and the maximum amplitude $\|v^N\|_\infty$. Once λ moves above 1, the amplitude grows rapidly for FTBS, CTCS, Lax–Friedrichs, and Lax–Wendroff, matching the theoretical stability boundary.

Sweep in λ: Unstable and Robust Schemes

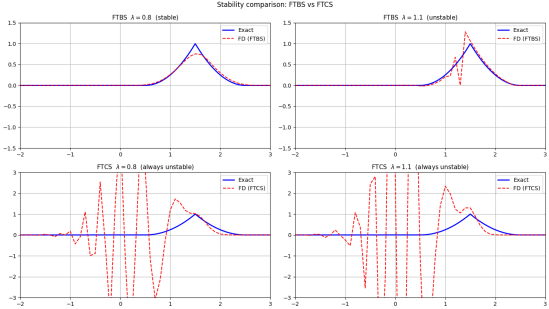


FTFS uses the wrong spatial bias for a right-moving wave, and FTCS is unstable for pure advection, so both show large amplitudes even when $\lambda < 1$. Beam–Warming remains bounded for the tested values because its stability range is $0 \le \lambda \le 2$. The implicit upwind/centered methods, BTBS, BTCS, and Crank–Nicolson, also remain bounded as λ increases, although BTBS and BTCS are visibly more diffusive.

Boundary and Stability Checks



For $T = 1.5$, the pulse support is approximately $[0.5, 2.5]$, so $b = 3$ is already far enough away. The errors for $b = 3, 4, 5$ coincide.



The direct comparison confirms the theory: FTBS is stable at $\lambda = 0.8$ and breaks down above the CFL limit, while FTCS is unstable for advection.

Conclusions

- The upwind direction matters: for $u_t + u_x = 0$, FTBS is the stable explicit upwind method; FTFS is downwind and unstable.
- FTCS is unstable for pure advection because $|G| > 1$ for nonzero Fourier modes.
- Lax–Friedrichs, Lax–Wendroff, and CTCS obey the expected CFL threshold $|\lambda| \leq 1$.
- Beam–Warming remains stable for the tested $\lambda > 1$ values, consistent with the wider interval $0 \leq \lambda \leq 2$.
- BTBS, BTCS, and Crank–Nicolson remain stable as λ increases, but stability alone does not imply high accuracy.
- The convergence plots agree with the theory after accounting for the nonsmooth corner in u_0.